Project report DM2905 - Incorporating challenges to influence users' teleportation performance in VR

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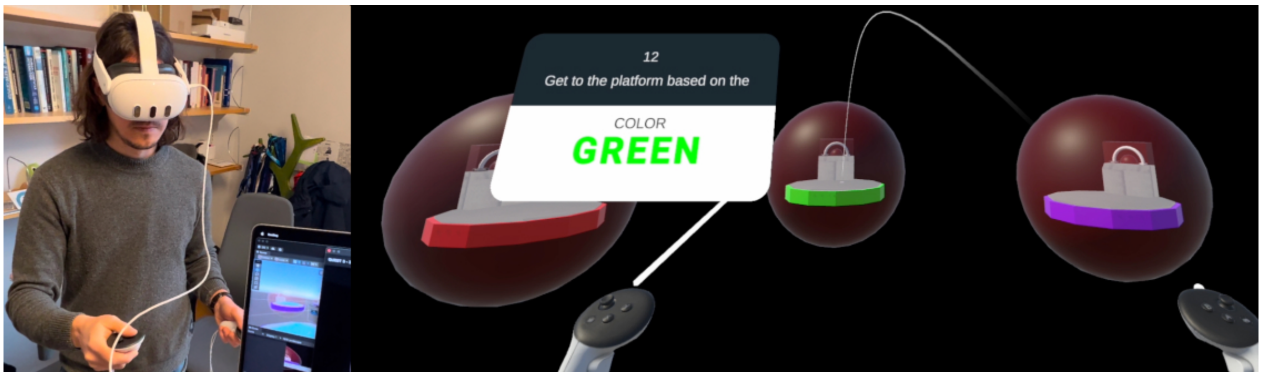

Figure 1: Researchers conducting an internal test to validate game stressors

Abstract

VR teleportation is a widely adopted technique for synthetic movement in virtual environments. It has been extensively evaluated and modified, as well as placed into game-like scenarios where users teleport to collect coins or find targets. Discussions in prior research highlight the interest in including challenges, in-game pressure, or placing the user in fast-paced scenarios during teleportation evaluations, suggesting that such contexts may yield more significant differences between methods [4, 10, 16]. Still, no clear examples of the type of challenges are provided or discussed. Building on these discussions, this work presents an implementation of two known teleportation techniques (Point & Teleport and Orientation adjustment), a set of three factors that affect the in-game challenge difficulty, and a study design for their evaluation, coupled with procedural guidelines. The study aims to address the research question: *What is the impact of incorporating in-game challenges on users' teleportation performance in VR environments?* Finally, this work outlines experimental considerations and addresses known issues for researchers interested in conducting the proposed study.

1 Introduction

VR locomotion techniques enable users to navigate virtual environments by controlling virtual avatars, eliminating the need for physical movement. Among these methods, teleportation [4] has been extensively evaluated against other techniques and various modifications, such as viewport orientation [10] or height elevation selection [23] at the destination. It is a widely used method in VR games, particularly for its ability to minimize virtual sickness and allow rapid traversal of virtual spaces.

A recurring topic in teleportation evaluation is the inclusion of in-game challenges or fast-paced scenarios, with discussions often suggesting that in-game pressures might significantly influence teleportation performance. However, clear guidelines or well-defined examples of the types of challenges that could be implemented remain open to interpretation.

This study addresses these discussions by designing a user study that incorporates specific in-game challenges to evaluate teleportation performance. These challenges include *Time Pressure*, where the available time to complete tasks is gradually reduced to induce stress; *Biased Instruction*, which introduces cognitive interference by distorting the task instructions; and *Attention Demand*, where the size of the teleportation destination platform shrinks dynamically to increase difficulty. By emulating a challenging gaming environment, this work seeks to assess how these factors impact user performance during VR teleportation tasks.

The main contributions of this work are the implementation of two known teleportation techniques (Point & Teleport and Orientation Adjustment), a documented process outlining the rationale for the selected challenges, and a study design for their evaluation, with procedural guidelines for future research.

1.1 Motivation

Previous authors have shown interest in adding challenges or applying in-game pressure to influence the use of teleportation, but the concrete examples of what kind of obstacles remains open to interpretation. Therefore, this project serves as a learning journey for advancing the understanding of VR teleportation within an environment that incorporates in-game challenges. It aligns with the following learning outcomes:

1. Critically appraise published VR teleportation and psycho-physiological literature and research from known conferences and proceedings in VR and CHI, such as I3D, CHI-PLAY, and VRST.
2. Identify stimuli that induce high levels of arousal in virtual environments.
3. Set up a virtual environment that employs different arousal stimuli to perform teleportation tasks.
4. Evaluate the user experience and performance of different teleportation methods in virtual environments that induce high levels of arousal using self-reported and psycho-physiological metrics.

1.2 Research question

This project aims to conceptualize the research instruments necessary for answering the following research question:

RQ1: What is the impact of incorporating in-game challenges on users' teleportation performance in VR environments?

2 Related work

Teleportation is a method of locomotion widely used in VR games. It is standard in today's games, particularly in first-person shooters where accuracy and speed are expected, such as in Hyper-Dash [7] and Doom VFR [21]. It was first introduced in 2016 as the Point & Teleport technique [4], and it has been extensively explored and evaluated. It offered a better option for fast scene traversal without physically moving and with less virtual sickness than other locomotion approaches, such as continuous movement control by joystick or head tilting [15]. However, path integration and spatial orientation became challenges for this technique, as it creates an instant change in the rendered image in the player's viewport [19]. This instantaneous image refresh motivated the exploration of viewport transitions, hyper-jumps, or dashes instead of instant changes to help improve the player's experience[3, 2]. Moreover, several variations of the technique have been introduced throughout the years and evaluated against their ability to maintain less virtual sickness and immersion by changes in several factors, such as the orientation at the destination, the height at the destination, or complete 3D teleportation in-mind air [10, 23, 16]. Although one can still see new studies around the impact of teleportation and its variation on player performance, companies like Meta offer the classic Point Teleport as their default locomotion feature in their SDK.

Previous studies have often discussed adding challenges or placing the user in fast-paced scenarios for future evaluations of teleportation techniques, arguing that this would yield significant differences in performance between methods [4, 10, 16]. Some have suggested applying in-game pressure or game mechanics to influence the use of teleportation, acknowledging that in-game situations might affect performance. Still, it is not clear what kind of game pressure or challenges these authors are referring to, and it is left relatively open to interpretation. Our work builds on such discussion around what would happen if the player were placed in a scenario with in-game pressure that presents a more challenging virtual environment for teleportation. But before proposing such a study, it is essential to clarify how we can materialize and implement concrete in-game pressure.

The literature on gameplay balancing introduces the concept of absolute difficulty, driven by intrinsic skill and stress, to shed light on what type of in-game pressure could influence the use of teleportation and, ultimately, the player's performance. In the context of game design, pressure is tightened to time and is measured by stress, which depends on how a player perceives the impact of a time constraint on the ability to meet a challenge [1]. The less time, the more stressful the situation. On the other hand, Intrinsic skill is the skill level required to meet a challenge under the assumption of an unlimited amount of time. These factors come together to define the absolute difficulty of an in-game challenge.

By understanding how to modify the difficulty of a challenge, there is a wide variety of factors that can be considered; we can influence the use of teleportation by affecting the skill level required to teleport, placing obstacles to overcome before teleporting, or influencing the time available to reach a destination. Several in-game variables, such as enemies' AI behavior, spawn rate, damage contribution, resource availability, visibility, health recovery rate, time span, and environment layout, have been manipulated in real games and lab setups to affect the game's perceived challenge [14, 17, 5, 9]. Also, cognitive challenges have been added by deriving from psychological tests [11]. In response to these changes, players have reported experiencing more stress, which has been validated through skin conductance measurements and standard questionnaires [6]. These previous works set a theoretical ground for our work to explore what kind of elements can be added to teleportation tests to make them feel more challenging for participants.

3 Methods and Techniques

Several sub-questions had to be answered to propose the user study design: What is the definition of stress in a game context? How can a scenario induce stress that feels like it belongs to a game context? Moreover, how can stress be measured?.

3.1 Defining stress in video games

Through a literature review of HCI, engineering databases, and medical resources, it can be found that stress is a state of a larger spectrum called arousal, ranging from activation to sleepiness. Russell et al. [20] define arousal as a continuum, and within that continuum, stress lies in an area that is considered activating. Although instrumental, for the game design literature, stress is attached to a time constraint that increases the absolute difficulty of a challenge. [1]. So, to converge into a definition, this work will consider time constraints to be a stressor that generates the perception of a task as being more difficult.

3.2 Manipulating the skill level

It is essential to recognize that choosing and including a stressing game mechanic in the study, representative of all possible options within VR fast-paced games, would have required a thorough literature review, as challenges in video games are diverse [1]. However, the perceived difficulty of a game challenge lies in the intrinsic level of skill required to meet the challenge. Therefore, as teleportation's main concern is to translate one's position, we decided to focus on increasing the intrinsic level of skill required to choose the next teleportation target by organically hindering the user's reaction time.

Gradl et al. [11] demonstrated that overriding automatic responses using the Stroop test in VR environments causes a delay in reaction time. In their experiment illustrated in Figure 2, people were asked to read the stoop card and shoot a ray to a wall with the corresponding color. This incongruency interferes with the automatic brain response to reading the word and the more taxing task of paying attention to the font's color. The phenomenon was first explored by Stroop[22], who coined the term cognitive interference. Stroop's powerful discovery can be extrapolated into teleportation scenarios, where the colored walls are teleportation targets.

Interestingly, Gradl et al.'s test conditions also included time constraints and environmental stressors; in the latter, they shrunk the size of the walls to increase the attention needed to aim at the correct wall. Wall size reduction builds on Fitt's Law [8], which establishes that reaction time depends on the distance to the target and the target's size. An environmental modification of such sort can be translated into shrinking teleportations' destinations, which targets gradually decrease.

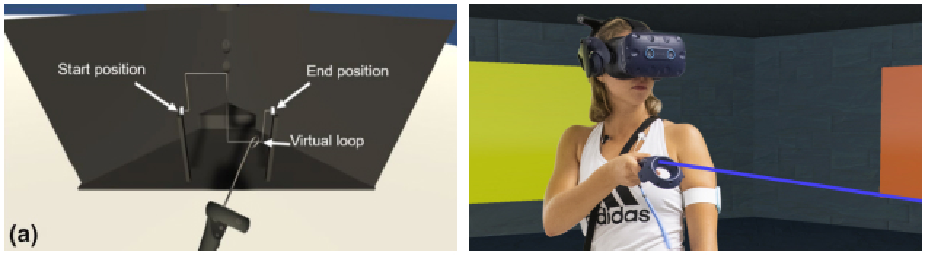

Figure 2: (Left) Radhakrishnan et al.'s VR training environment. (Right) A participant in Gradl et al. Stroop room test, pointing at walls with colors

3.3 Incorporating Game-like feeling

Lastly, building on the idea of a game-like scenario, a key component added to the experience was to give a purpose to the teleportation task; with that in mind, a subsequent action had to be added after teleportation before moving to the next teleportation target. Previous works have proposed recollection tasks such as collecting coins [16]. However, the former identified that participants started to be less accurate in their recollection, looking to complete the trial in less time. Therefore, an interest in secondary tasks that organically induce attention drove this research toward Radhakrishnan's [18] work, which took the buzz-wire game to develop motor skill training in VR as shown in Figure 2. Although not directly categorized as an attention- or stress-inducing element, the task inherently demanded fine motor control and precision, potentially increasing cognitive load and shifting focus from speed to controlled, deliberate action. Adding the buzz-wire mechanic, as depicted in Figure 3, to induce cognitive demand is expected to add both complexity and meaning, encouraging participants to be mindful of their actions before pursuing the next target.

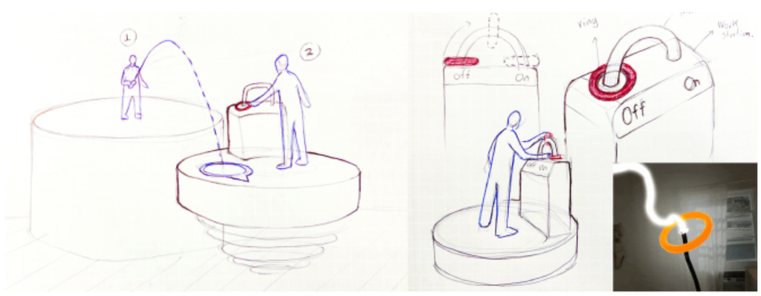

Figure 3: Initial sketch of the buzz-wire interaction where the player would have to move a ring from point A to point B without touching the arc, thus activating the platform.

3.4 Evaluating stress

To evaluate the user's arousal state, this study adopts Gradl et al.'s use of the Short Stress Questionnaire for self-reported stress levels alongside electrodermal activity (EDA) sensors for physiological validation. Task difficulty is assessed using the NASA Task Load Index (NASA-TLX), following Weissker [23]. Together, these metrics provide a comprehensive framework for understanding user performance and stress responses when incorporating in-game challenges.

Although Gradl et al.'s and Radhakrishnan et al.'s work is not strictly targeted to game development or teleportation, both have proven results of their stimuli being stressful and cognitively demanding within VR. Both papers provided valuable grounding for this user study's design, which will be presented in the following section.

4 Results, Implementation and Evaluation

This work outlines and iteratively develops a tool tested internally by the researcher to refine the stimuli and experimental design. The framework detailed herein provides a foundation for future research by offering a prototype, questionnaire, and procedural guidelines to facilitate subsequent evaluations. The experiment is designed to investigate how exposure to in-game challenges, in which intrinsic skill level has been modified, further explained in the upcoming section, affects teleportation performance, perceived stress, and workload in VR environments. The source code for the proposed tool can be found in the project's GitHub repository[1].

In this experimental design, the subjects have to teleport between 6 hexagon-layout platforms as seen in Figure 4. Each platform represented a different color (red, blue, green, yellow, orange, and purple). Subjects are given one teleportation instruction at a time in the form of color displayed on a VR GUI to move to a destination. Once at the destination platform, they activate a mechanism to cue the system for the next platform.

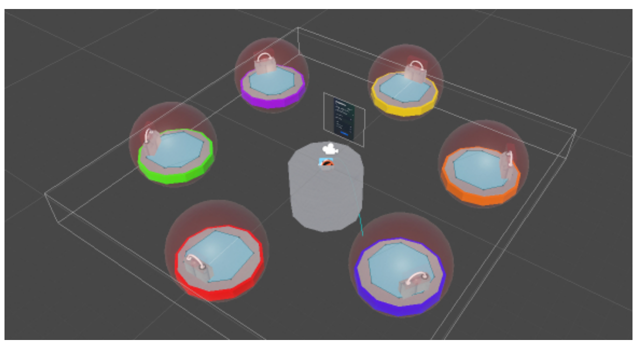

Figure 4: Hexagonal layout of the StressPort scene, with one color per platform.

4.1 Participants

We recommend conducting this study with at least 16 to 32, participants, as this study has eight conditions, explained in the following section. This will allow the experimenter to create a balanced Latin square [2] and reduce learning curves due to ordering effects. To keep a similar age sample as the previous works that have evaluated teleportation [4, 10, 16], we recommend choosing individuals between 20 and 35 years old, with prior experience in VR experiences (at least 5 hours, as in [16]). Questionnaires that aim to delve into participants' individual differences with VR can be added. Lastly, as in [11] study, all participants must have normal or corrected-to-normal vision, and it should exclude color-blind individuals as this study depends on color-based tasks.

[1] https://github.com/DavidGiraldoCode/s-Evaluating_VR_teleportation_under_stressful_gameplay

[2] https://damienmasson.com/tools/latin_square/

4.2 Study Design

This within-subjects experiment includes two independent variables: (1) the teleportation method and (2) the difficulty multiplier: **Teleportation Method (TM):** In the present prototype, participants can use two types of teleportation: (a) Point&Teleport [4], where orientation remains unchanged, and (b) teleportation with orientation control, where the user defines the orientation at the destination before teleportation [10]. **Difficulty Multiplier (DM):** The experiment utilizes three stimuli that influence the absolute difficulty of the teleportation task: *Time Pressure (TP):* The time to meet a task gradually decreases to induce stress. *Biased Instruction (BI):* The instruction to teleport to a platform is distorted (Cognitively Interfered Instruction) by presenting a color word in mismatched ink color, requiring the user to teleport based on font color rather than word content. *Attention Demand (AD):* The third multiplier consists of the platform shrinking toward a centered pivot place at the center top surface. This means that the available teleportation area reduces, while the center of the platform never moves.

These Difficulty Multipliers are expected to delay reaction times and increase perceived stress and cognitive load, thereby impacting teleportation decision-making. To compare the delta in teleportation performance, Table 1 organizes the variables in four tuples; the participants will go through base conditions that replicate a traditional locomotion evaluation with any stressing stimuli. Then, the game stressors are gradually added.

Table 1: Conditions distribution, gradually adding the stressors. This distribution can be changed based on the researcher's requirements.

Condition	**TM**	**DM**
A base	Point&Teleport	None
B base	Orientation	None
A1	Point&Teleport	Time Pressure
B1	Orientation	Time Pressure
A2	Point&Teleport	Biased Instruction
B2	Orientation	Biased Instruction
A3	Point&Teleport	Attention Demand
B3	Orientation	Attention Demand

4.3 Task

In this experiment, participants are tasked with navigating between colored platforms using teleportation, aiming to activate a mechanism on each platform to unlock subsequent instructions. Participants will start on the red platform, as in Figure 5, and perform the following actions:

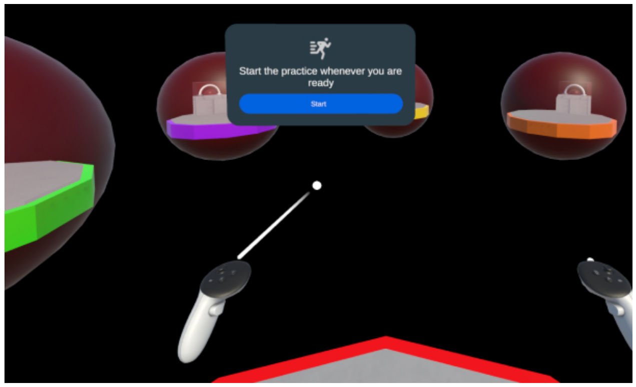

Figure 5: Starting point on the red platform. The participant is about to start the practice round.

(1) Identification and Search: Participants read a prompt on the left-hand GUI, as depicted in Figure 6, directing them to a specific target platform. Then, they located the designated platform based on the prompt.

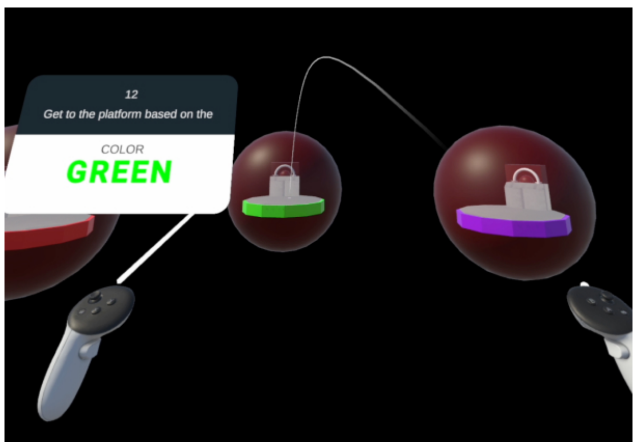

Figure 6: The wrist-mounted GUI on the participant's left-hand prompts the color and displays the timer. With the right hand, they aim at the platform, press the trigger, and release to teleport.

(2) Navigation: Participants teleported to the specified platform. The hexagonal platform arrangement allows participants to teleport directly to any target without intermediary stops. This layout ensured seamless movement between platforms, maintaining a dynamic and uninterrupted gameplay flow. Incorrect platform selections triggered an error sound to provide immediate feedback, while successful activations produced a confirmation sound. **(3) Focused Action:** Upon arrival, they engaged with a mechanism on the platform to progress to the next instruction. The mechanism for unlocking subsequent instructions, as shown in Figure 7, drew inspiration from Radhakrishnan et al.'s work on motor skill training in virtual reality. This component introduced an additional layer of challenge, demanding fine motor control and precision, thereby enhancing the cognitive and physical demands of the task. These steps were designed to emulate typical gameplay challenges, requiring coordination, reaction time, precision, and accuracy, aligning with Adams' principles outlined in Fundamentals of Game Design [1].

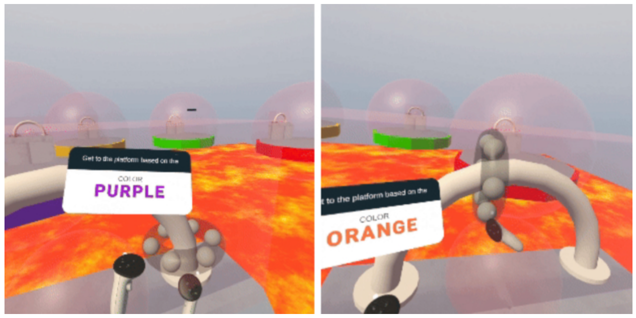

Figure 7: The buzz-wire interaction makes the player shift into a mindful state of attention to activate the platform. Touching the path resets the ring to its initial position.

4.4 Apparatus

The experimental software was developed on the Meta Quest 3 VR headset using Meta's Interaction SDK in conjunction with Unity's OpenXR SDK and its Built-in Render Pipeline. The application is installed directly onto the HMD for a wireless experience, and the viewport must be streamed to a MacBook Pro via the Meta Quest Developer Hub for real-time monitoring. More implementation details can be found at the GitHub repository[3].

For low-cost physiological data recollection, this study recommends using the BITalino (r)evolution Board BT/BLE to measure Electrodermal Activity (EDA). Sensors can be placed on participants' non-dominant hand Figure 8. At the same time, all interactions (teleportation, rotation, and grabbing) are conducted with the dominant hand to standardize the input method across participants. However, changes in the source code would be required to account for left-handedness.

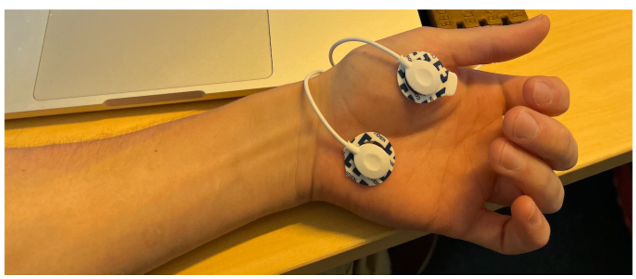

Figure 8: Recommended placement of the EDA sensors on the non-dominant hand.

4.5 Measures

To measure the performance delta between teleportation methods, we track several dependent variables. (1) *Perceived workload* is assessed using the NASA-Task Load Index [12], providing insights into participants' cognitive and physical effort. (2) *Perceived stress* is measured with the Short Stress State Questionnaire [13], which evaluates participants' stress levels during gameplay.

Quantitative data is recorded through various metrics. (3) *Average time at platform* captures the time in seconds spent at the target platform for activation. (4)*Reaction time* measures the time

[3] `https://github.com/DavidGiraldoCode/s-Evaluating_VR_teleportation_under_stressful_gameplay`

in milliseconds between receiving the prompt and initiating the first teleportation. (5) *Average orientation delta* records the absolute change in player orientation, in degrees, from the starting to the final view after teleportation.

Additional error and precision metrics are tracked. (6) *Average wrong platform given a color*, counts instances where the player landed on the incorrect platform after receiving a color prompt, and (7) *Buzzwire touches at platform* counts the number of buzz wire touches during platform activation attempts.

4.6 Procedure

Before the Study Execution, participants are welcomed to the experiment, asked to provide signed consent for data protection and complete a pre-study questionnaire to assess baseline stress levels. After the initial preparation, each participant is assigned a unique session ID and equipped with an Electrodermal Activity (EDA) device on their non-dominant hand. This setup ensured that all interactive tasks during the experiment, such as teleportation and item manipulation, were consistently managed using the dominant hand. *During the Study Execution*, the experiment must be structured according to a Balanced Latin Square design, allowing for randomized condition distribution to mitigate any learning effects across the teleportation tasks. Participants begin with a teleportation practice phase lasting approximately two minutes, giving them an opportunity to familiarize themselves with the mechanics. Following this practice, they proceed to the main trial phase, which includes 30 teleportation tasks over five minutes. Data collection is integrated into the software at specified intervals throughout. EDA measurements must be recorded at the beginning, during, and at the conclusion of each task phase. After each condition, participants complete a post-condition questionnaire to evaluate their perceived stress using the Short Stress State Questionnaire [13], and workload using the Nasa-Task Load Index [12]. Upon the conclusion of each session, experimenters should verify the generation of .csv data files on the headset, confirm the EDA data recordings, and save responses in preparation for the next condition. *After the Study Execution*, participants will complete an additional questionnaire focused on preferences and evaluations related to teleportation methods and orientation features. They will note which condition was associated with greater perceived stress and provide justifications for their preferences. Participants then rank the impact of the different difficulty multipliers on their subjective experience, offering reasons for their choices. After this final feedback, experimenters should download the .csv data from the HMD, retrieve the EDA recordings, and provide a debrief to each participant to conclude the study.

4.7 Internal evaluation

A walk-through of the prototype was conducted internally, validating assumptions about the difficulty multipliers, as shown in Figure 9. Time constraints were effective in heightening alertness, while Stroop stimuli successfully induced cognitive interference. However, the subject quickly adapted by ignoring text and focusing on colors. To counteract this learning effect, dynamic rule changes, as used in the StroopRoom [11] experiment, could be implemented. Shrinking platforms to induce attention demand underperformed due to their slow shrinking rate and shrinking limit. Nonetheless, heightened alertness was reported when looking down during platform shrinkage, suggesting the need for modification or possible removal of this stimulus. Testing the prototype also revealed limitations in feedback mechanisms, particularly for color prompts, which the test subject often missed while focusing on the buzz-wire task. This highlighted the importance of adding auditory or haptic feedback, as GUI prompts attached to the left controller were not always visible in the viewport, as it can be see in . These insights suggest key areas for future refinement.

Figure 9: Researchers testing the prototype in the lab. The buzz-wire turns red when the ring touches it.

5 Discussion

Developing experimental prototypes differs significantly from traditional software development, emphasizing rapid iteration and flexibility. In the first iterations of the prototype, the experimental software consisted of a robust system with predefined conditions and automated resets, structured similarly to a level-based game as shown in Figure 10.

However, this approach limited adaptability for experimental adjustments. Changing and adding new independent variables rapidly became a bottleneck. After careful consideration, the prototype transitioned to a more flexible settings menu, shown in Figure 10, allowing the toggling of independent variables (Teleportation Methods and Difficulty Multiplier). This change significantly improved the implementation's versatility and usability for experimentation.

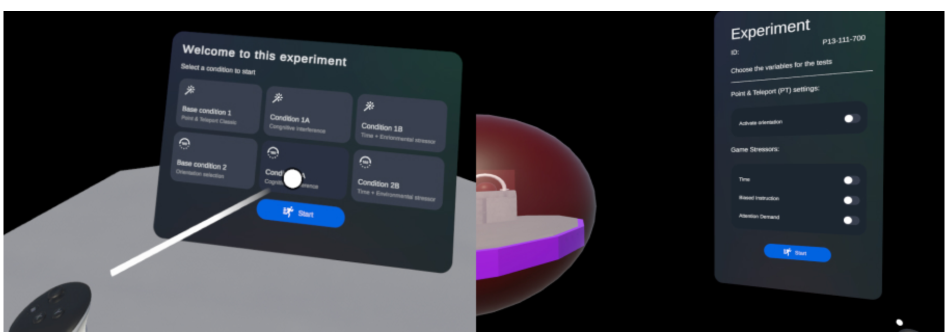

Figure 10: (Left) The initial condition menu was inspired by a "choose your own adventure" type of interaction and level selection. (Right) The final experiment setting menu enables the toggling of difficulty multipliers based on the conditions to test.

Incorporating EDA sensors posed several challenges; previous work informed that EDA is deemed to include noise in the data due to variations in participants' skin types (e.g., dry, sweaty, or oily) and difficulties in maintaining consistent placement across participants of different body sizes.

To address these issues, EDA is proposed as a supplementary measurement, with the Short Stress Test questionnaire and NASA workload assessments serving as the primary tools for evaluating arousal. Additionally, integrating EDA sensors required significant adjustments to the interaction system. The proposed study design suggests that the sensors should be placed on participants' non-dominant hands to maintain measurement consistency, which limits VR interaction. Consequently, teleportation and grabbing inputs were consolidated on the right hand only.

6 Learning reflection

This project has significantly strengthened my software development skills and research acuity while fostering a newfound growth mindset. Due to time constraints, skill limitations, and growing implementation complexity, the project scope had to pivot, as agreed upon with my supervisor. Despite this, the learning objectives were addressed to the greatest extent possible. The experimental protocol was fully outlined, including a comprehensive step-by-step procedure and accompanying questionnaires, and the prototype's source code was meticulously documented for future work. The focus shifted toward designing and proposing an experimental framework, providing theoretical grounding, delivering a functional prototype, and presenting a user study design to guide subsequent research efforts.

6.1 Reviewing Learning Objectives

With the updated scope in mind, let us revisit the Learning Outcome (LO) to assess how they evolved from the original proposal. This reflection highlights achievements, strengths, weaknesses, and areas for future improvement.

6.1.1 LO1 - *Critically appraise published VR teleportation and psychophysiological literature from renowned conferences such as I3D, CHI-PLAY, and VRST*: The literature review was thorough and focused on foundational VR teleportation methods, such as "Point and Teleport" [4] while integrating standard evaluation tools like the NASA Task Load Index (NASA-TLX)[12]. ACM DL and IEEE served as primary resources, supplemented by broader databases such as the American Psychological Association and Springer, which were instrumental in exploring the physiological dimensions of arousal. Notable studies, including the Stroop Room [11] tests, played a critical role in shaping the methodology, as they gave us a stressing stimulus to start with. Additionally, adopting Zotero for source organization facilitated efficient sharing and collaboration with my supervisor. As the research progressed, I refined my search strategies, employing targeted keywords and advanced queries in IEEE, ACM, and Google Scholar. A significant strength was anchoring the experimental design on established work, particularly [11], which provided a robust foundation for implementing effective difficulty multipliers.

6.1.2 LO2 - *Identify stimuli that induce high levels of arousal in virtual environments*: Difficulty multipliers were carefully grounded in prior literature, including insights from game design texts and research studies. This was the most challenging part of the project, as choosing how to hinder users' usage of teleportation had to be consequent with the task.

Defining and contextualizing *arousal* proved to be a significant challenge. Initially, I relied on psychological and medical literature to understand arousal as a spectrum rather than a binary condition. Fortunately, incorporating concepts from Fundamentals of Game Design [1] helped bridge clinical definitions with practical applications, offering a more gaming-centric perspective on how users perceive challenge and difficulty and how this is related to stress perception. This combined grounding revealed that not all high-arousal states were ideal for this project. In the end, the concept of intrinsic skill, time-based pressure, and absolute difficulty provided by Adams helped me narrow down the scope. Selecting the Stroop test, as recommended by [11], proved to be a suitable choice for affecting the intrinsic level of skill required for teleportation.

If I were to revisit this objective, I would prioritize consulting design theory books to establish a foundational understanding before diving into more complex clinical definitions. This approach would streamline the process and mitigate the challenges of translating intricate medical definitions into HCI implementations.

6.1.3 LO3 - *Set up a virtual environment employing various arousal stimuli for teleportation tasks*: I learned to prioritize flexibility and variable manipulation when developing experimental prototypes, as they differ significantly from traditional software development. The implementation phase began with Unreal Engine and Meta's SDK on macOS. However, complications with SDK updates and compatibility led to a shift to Unity, which proved to be a much better fit for the workflow. Unity's development environment offered accessibility and clear documentation, enabling faster progress. Setting up VR capabilities for MacOS required multiple dependencies, including Android Bridge, Meta Quest Developer Hub, and a VR virtual simulator powered by Vulkan's API. While the initial setup presented challenges, Unity provided a more streamlined experience. Navigating the complex input systems of Head-Mounted Displays (HMDs) required substantial time, particularly for customizing teleportation mechanics. A scalable architecture was developed, initially enabling the implementation of most features without the HMD, which ultimately helped to refine VR-specific interactions like teleportation and object grabbing.

The experimental prototype reached approximately 90% completion, with two key elements remaining: implementing data logging into a CSV file and resolving a bug affecting the teleportation orientation feature, which inadvertently disabled object grabbing when both functionalities were active in the same scene. If I were to do this project again, I would define a smaller feature list and have explored and studied Unity's VR SDK much sooner.

6.1.4 LO4 - *Evaluate user experience and performance of different teleportation methods in high-arousal virtual environments using psycho-physiological metrics such as electrodermal activity (EDA)*: We initially planned for an exhaustive user evaluation with 24 participants, a careful screening procedure, and consequent data analysis. Still, later in the process, I noticed that this expectation was falling out of scope. Therefore, we focused on honing the intersection we had so far and validated the experience through a demo test using my supervisor as the subject of study. I also created a detailed experimental script, including step-by-step instructions and a precise timetable, which is included in the appendix. My supervisor introduced me to the Latin Square design, a method for randomizing conditions to minimize bias and account for learning effects. This preparation streamlined the implementation process and clarified priorities, enabling us to simplify certain steps and focus on the study's critical aspects.

7 Conclusion

An experimental study design was presented alongside the necessary tools to evaluate VR teleportation when including in-game challenges. This instance of the prototype includes two teleportation methods and three difficulty multipliers, but further methods and multipliers can be added according to the research's needs. Also, the discussion underscored the importance of iterative design testing to gauge multipliers' efficacy. While the literature provided foundational insights, the implementation and hands-on testing revealed nuances that were not initially apparent. Sound feedback, dynamic Stroop rules, and adjustments to attention-demanding stimuli are critical next steps for enhancing the prototype's functionality. The design hopes to help advance the understanding of VR teleportation evaluation when including in-game challenges and serve as a starting point for future researchers and VR game developers.

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8 Appendix

8.1 Experimental script

The following describes the procedural guidelines for conducting the user study. **Previous to the test**, the researcher will:

1. Welcome participant
2. Ask the participant to sign the data protection form
3. Ask them to fill in the pre-study questions to control their initial level of stress

During the test: The participant will go through the conditions in the order assigned by the Balanced Latin Square. The researcher will:

1. Input the participant's ID in the HMD session.

2. Set up the EDA device on the non-dominant hand. *For this to happen, the dominant hand will have to do all the work (teleport, rotate, and grab).

3. Set up the subject with the HeadSet (1 min) and ask them to toggle the variable value for the corresponding condition.

   (a) Tick the EDA device once

4. Run the practice task with six teleportations (2 min)

   (a) Tick the EDA device two times

5. Run the trial tasks with 30 teleportations (5 min)

   (a) Tick the EDA device three times

6. Remove headset (do not remove EDA)

   (a) If it is the last condition, remove EDA.
   (b) Clean HeadSet

7. Hand in the post-condition questionnaire, which comprises the Stress questionnaire and the workload evaluation.

8. While participants answer, validate that the .cvs file has been created in the headset and that EDA data has been recorded.

9. Save the answers

10. Move to the next condition if so. Go to step 3.

After the test: research will make questions about the experience in retrospect using the Post-study questions (further explained below). Thank the participant for the time. Finally, download the .csv file from HMD and the EDA data.

8.1.1 Pre-study questions:

1. How old are you? (type only the number)

2. You identify as: (Male, Female, Prefer not to say)

3. What kind of previous VR experience do you have? (explain briefly)

4. How much have you played VR shooter video games, specifically? (Have not played, At least once, More than once)

8.1.2 Post-study questions:

1. What type of teleportation leads you to feel more stressed? (Tradicional, without orientation, With orientation).
2. What made that teleportation more stressing for you? (please elaborate on your answer)
3. What stressor had higher influence in your overall feeling of stress during the experiment? (Cognitive (mismatch of text and colors), Shrinking platforms)
4. What made that stressor more stressing for you? (please elaborate on your answer)
5. Would you like to share any other comments? (Write NA if not)

8.1.3 Stress questionnaire: Please indicate how well each word describes how you feel at the moment. Mark 1 (less agree) or 5 (more agree).

- I feel dissatisfied. (1-5)
- I feel alert. (1-5)
- I feel depressed. (1-5)
- I feel sad. (1-5)
- I feel active. (1-5)
- I feel impatient. (1-5)
- I feel annoyed. (1-5)
- I feel angry. (1-5)
- I feel irritated. (1-5)
- I feel grouchy. (1-5)
- I am committed to attaining my performance goals. (1-5)
- I want to succeed on the task. (1-5)
- I am motivated to do the task. (1-5)
- I'm trying to figure myself out. (1-5)
- I'm reflecting about myself. (1-5)
- I'm daydreaming about myself. (1-5)
- I feel confident about my abilities. (1-5)
- I feel self-conscious. (1-5)

- I am worried about what other people think of me. (1-5)
- I feel concerned about the impression I am making. (1-5)
- I expect to perform proficiently on this task. (1-5)
- Generally, I feel in control of things. (1-5)
- I thought about how others have done on this task. (1-5)
- I thought about how I would feel if I were told how I performed. (1-5)

Based on the Short Stress State Questionnaire [13].

8.1.4 Workload evaluation: We are interested in the workload you experienced while completing this task. As several different factors can cause workload, we ask you to rate each factor individually on the scales provided. Note that one means very low and 20 is very high, except for the last question about Performance.

- Mental Demand: How mentally demanding was the task? (1-20)
- Physical Demand: How physically demanding was the task? (1-20)
- Temporal Demand: How hurried or rushed was the pace of the task? (1-20)
- Effort: How hard did you have to work to accomplish your level of performance? (1-20)
- Frustration: How insecure, discouraged, irritated, stressed, and annoyed were you? (1-20)
- Performance: How successful were you in accomplishing what you were asked to do? *note: Performance is reversed, goes from 1 point being good to bad being 20 points. (1-20)

Based on the NASA-Task Load Index [12].